RAJARATA UNIVERSITY OF SRI LANKA • DEPARTMENT OF COMPUTING

FixSphere — Video Presentation Script & Dubbing Guide

ICT 1306: Object Oriented Programming • 5-Minute Live Application Demonstration

Target Duration: Exactly 5 Minutes (00:00 – 05:00)

Platform: YouTube (Upload as Unlisted)

Repository: github.com/Ravindu2k4/Fixsphere

Technology: C++/CLI (WinForms) & MySQL 8.0

Team Members:
1. K. R. M. De Silva (6675 | ICT/2024/063) • 2. S. P. Abewickrama (6618 | ICT/2024/003) • 3. D. L. C. A. Liyanawaduge (6685 | ICT/2024/078)
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Section 1: Introduction & Business Scenario — 00:00 – 00:40 (40 sec)

🖥 VISUAL ACTION (WHAT TO SHOW ON SCREEN)

Display the Title Slide showing "FixSphere — Mobile Phone Repair Center Management System", group member names, index numbers, and the Rajarata University logo. After 15 seconds, switch screen to show the solution open in Visual Studio.

🎤 EXACT DUBBING SPEECH (READ ALOUD CLEARLY)

"Hello everyone, and welcome to our demonstration of **FixSphere**, a desktop management system designed for mobile phone repair centers.

This project was developed for module **ICT 1306: Object-Oriented Programming** at the **Rajarata University of Sri Lanka**.

In a typical repair shop, managing device intake, technician assignments, repair statuses, and billing can quickly become disorganized when relying on manual records. FixSphere solves this by delivering a centralized, role-secured Windows desktop application powered by Object-Oriented Programming principles and a relational MySQL database."

Section 2: Architecture & The 6 OOP Pillars Walkthrough

00:40 – 01:30 (50 sec)

🖥️ VISUAL ACTION (WHAT TO SHOW ON SCREEN)

In Visual Studio, open and scroll through `Models.h`, highlighting the classes `Person`, `Customer`, and `Employee`. Briefly switch tab to `DatabaseService.h`.

🎤 EXACT DUBBING SPEECH (READ ALOUD CLEARLY)

"Before running the live system, let us examine how our codebase demonstrates all six fundamental OOP pillars.

First, **Classes and Objects**: We modeled real-world entities into modular classes such as `Person`, `Customer`, and `Employee`, which are dynamically instantiated during execution.

Second, **Encapsulation**: Private attributes like names, contact numbers, and passwords are fully protected and accessible only via controlled public properties.

Third, **Abstraction**: Our base class `Person` is an abstract class with a pure virtual method, `GetRoleDescription()`, while our `DatabaseService` abstracts away MySQL query connections from the user interface.

Fourth, **Inheritance**: Both `Customer` and `Employee` inherit directly from our abstract `Person` base class to share core attributes without code duplication.

Finally, **Polymorphism**: Both derived classes provide distinct overrides for `GetRoleDescription()` and `GetSummary()`, resolved dynamically at runtime."

Section 3: Live Application Demo — Authentication & Roles

01:30 – 02:30 (60 sec)

🖥 VISUAL ACTION (WHAT TO SHOW ON SCREEN)

Launch `FixSphere.exe`. Show the clean `LoginForm` ("FixSphere - Employee Sign In"). Enter username `admin` and password `admin123`. Click **Sign In**. The main dashboard appears showing full sidebar navigation (Dashboard, Customers, Jobs, Employees, Reports).

🎤 EXACT DUBBING SPEECH (READ ALOUD CLEARLY)

"Now, let us run FixSphere and observe the live application.

Here is the login screen. FixSphere implements a strict three-tier Role-Based Access Control system supporting **Admin**, **Checker**, and **Repairer**.

We will begin by signing in as the **Administrator**.

Notice the responsive executive dashboard. As Admin, all navigation modules are accessible: the Dashboard overview, Customer Management, Repair Jobs, Staff Administration, and Revenue Reports. The Administrator has full system authority to assign jobs, edit technician allocations, and delete records."

Section 4: Live Core Workflow — Customer & Job Intake CRUD

02:30 – 03:45 (75 sec)

🖥 VISUAL ACTION (WHAT TO SHOW ON SCREEN)

1. Click **Customers** in sidebar. Click a row or enter a new customer (e.g., Kamal Perera, 0771234567).
2. Click **Repair Jobs**. Select the customer from dropdown, type Phone Model "Samsung Galaxy S22", issue "Cracked AMOLED Display", initial status "Pending". In the Repairer dropdown, select "Nimal Silva". Click **Save Job**. Point mouse to the newly added row in DataGridView.

🎤 EXACT DUBBING SPEECH (READ ALOUD CLEARLY)

"Now, let us demonstrate our live database CRUD operations.

Under the **Customers** tab, we can register new clients, update contact information, or search existing clients in real time. Every operation performs parameterized queries against our MySQL database.

Next, we open **Repair Jobs** to create a new intake. We select our registered customer, specify the phone model — for example, 'Samsung Galaxy S22' — and describe the diagnostic fault as 'Cracked AMOLED display'.

The initial status defaults to **Pending**. Because we are authenticated as Admin, the technician assignment dropdown is enabled. We assign this job to our technician, Nimal Silva, and save. The record updates instantly in our searchable DataGridView."

Section 5: Role Security, Financials in LKR & Reports

03:45 – 04:30 (45 sec)

🖥 VISUAL ACTION (WHAT TO SHOW ON SCREEN)

1. Click **Logout**. Log in as `repairer` (password `repair123`).
2. Show that Employee and Customer management forms are restricted/hidden.
3. Open **Repair Jobs**: select the assigned job, update status to **Finished**, set Cost to `15000`, click **Update Job**.
4. Log back in as Admin and show **Reports** tab: dynamic total calculated as `LKR 15,000.00`. Click **Export CSV**.

🎤 EXACT DUBBING SPEECH (READ ALOUD CLEARLY)

"Next, let us demonstrate role security. We log out and sign in as our **Repairer**, Nimal Silva.

Notice that administrative menus are automatically locked. Under Repair Jobs, the technician can only view their own assigned jobs and cannot reassign tickets to others.

The technician marks the repair as **Finished** and enters the servicing cost.

Notice that all monetary amounts throughout FixSphere are strictly formatted in **Sri Lankan Rupees (LKR)**.

In our **Reports** view, monthly revenue automatically aggregates only Finished jobs in LKR, excluding pending or unrepairable devices, with one-click CSV export."

Section 6: GitHub Verification & Conclusion

04:30 – 05:00 (30 sec)

🖥 VISUAL ACTION (WHAT TO SHOW ON SCREEN)

Switch to web browser showing the public GitHub repository at `https://github.com/Ravindu2k4/Fixsphere`. Show the repository root, `Models.h`, `setup_database.sql`, and `README.md`.

🎤 EXACT DUBBING SPEECH (READ ALOUD CLEARLY)

"Finally, here is our public GitHub repository at `github.com/Ravindu2k4/Fixsphere`.

The repository hosts our complete source code, Visual Studio project files, database schema script, and clear setup documentation.

In conclusion, FixSphere satisfies all assignment criteria: a unique business scenario, complete demonstration of all six OOP concepts, a multi-screen Windows desktop GUI, full MySQL CRUD operations, and strict role permissions.

Thank you for your time and attention!"

💡 Recording & Dubbing Tips for Best Marks

- **Keep Pacing Steady:** Practice reading along with the timestamps so your total video length stays between **4 minutes 45 seconds** and **5 minutes 00 seconds**.
- **Resolution:** Record in 1080p (1920x1080) using OBS Studio or Windows Game Bar (`Win + G`).
- **YouTube Privacy:** When uploading to YouTube, choose **Unlisted** (anyone with the link can watch, but it won't appear on public search).